
Approach

For this project, I developed a lightweight binary image classifier to distinguish between real photographs and photographs of digital screens (recaptures). The model is based on MobileNetV3-Large, a pretrained convolutional neural network optimized for mobile and edge devices.

I collected a custom dataset consisting of approximately 100 images, evenly split between the two classes. The dataset includes different lighting conditions, viewing angles, screen types, and real-world objects to improve robustness. During training, transfer learning was used by replacing the original classifier with a custom binary classification head while initially freezing the backbone. The final layers were then fine-tuned using the custom dataset. Data augmentation techniques such as random crops, horizontal flips, rotations, and color jitter were applied to improve generalization.

The trained model was integrated into a Flask web application that supports both image upload and live webcam capture. The application performs real-time inference and displays the predicted class, confidence score, and fraud probability.

Results

- * Model: MobileNetV3-Large (Transfer Learning)
- * Dataset: ~100 custom images (Real vs Screen)
- * Classification Accuracy: 93.75% on the held-out test set
- * Output: Fraud probability between 0 (Real Photo) and 1 (Photo of a Screen)

Future Improvements

With additional time, I would collect a significantly larger and more diverse dataset containing different devices, display technologies (LCD, OLED), brightness levels, reflections, and environmental conditions. I would also explore frequency-domain features, glare detection, and lightweight model quantization to further improve robustness while reducing inference time for deployment on mobile devices.